BEING SMOOTH IS SMOOTH LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's compressed instruction to combine four named results; it is delivered verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The general criterion for a property of morphisms to be local on the source. Same chapter, Lemma lemma-properties-morphisms-local-source, invoked at proof line 7151. Reduces the claim to four clauses.
- Zariski locality on source and target and preservation under precomposition for smooth morphisms. Morphisms, Lemmas morphisms-lemma-smooth-characterize and morphisms-lemma-composition-smooth, invoked at proof lines 7152-7154. Supply clauses (1)-(3).
- Permanence of smoothness under a smooth surjection on the source. Same chapter, Lemma lemma-syntomic-smooth-etale-permanence, invoked at proof line 7155. Supplies clause (4).

Source reference. Original lemma 036H: sources/stacks-descent/pinned-original-source.tex, source lines 6920–6920. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KM: sources/stacks-descent/pinned-original-source.tex, source lines 4640–4640. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01VA: sources/stacks-chow/tags.txt, source lines 2439–2439. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01V6: sources/stacks-chow/tags.txt, source lines 2435–2435. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is smooth" is smooth local on the source.*

Proof. Combine Lemma 036H with Morphisms, Lemma 01V6 (local for Zariski on source and target), Morphisms, Lemma 01VA (pre-composing), and Lemma 02KM (part (4)). □